

OpenStax Calculus Volume 2

Section 6.4 Exercises: Working with Taylor Series

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In the following exercises, use appropriate substitutions to write down the Maclaurin series for the given binomial.

1. $(1 - x)^{1/3}$
2. $(1 + x^2)^{-1/3}$
3. $(1 - x)^{1.01}$
4. $(1 - 2x)^{2/3}$

In the following exercises, use the substitution $(b + x)^r = (b + a)^r(1 + \frac{x-a}{b+a})^r$ in the binomial expansion to find the Taylor series of each function with the given center.

5. $\sqrt{x + 2}$ at $a = 0$
6. $\sqrt{x^2 + 2}$ at $a = 0$
7. $\sqrt{x + 2}$ at $a = 1$
8. $\sqrt{2x - x^2}$ at $a = 1$ (Hint: $2x - x^2 = 1 - (x - 1)^2$)
9. $(x - 8)^{1/3}$ at $a = 9$
10. $\sqrt{x}$ at $a = 4$
11. $x^{1/3}$ at $a = 27$
12. $\sqrt{x}$ at $a = 9$

In the following exercises, use the binomial theorem to estimate each number, computing enough terms to obtain an estimate accurate to an error of at most $1/1000$.

13. [T] $(15)^{1/4}$ using $(16 - x)^{1/4}$
14. [T] $(1001)^{1/3}$ using $(1000 + x)^{1/3}$

In the following exercises, use the binomial approximation $\sqrt{1 - x} \approx 1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16} - \frac{5x^4}{128} - \frac{7x^5}{256}$ for $|x| < 1$ to approximate each number. Compare this value to the value given by a scientific calculator.

15. [T] $\frac{1}{\sqrt{2}}$ using $x = \frac{1}{2}$ in $(1-x)^{1/2}$
16. [T] $\sqrt{5} = 5 \times \frac{1}{\sqrt{5}}$ using $x = \frac{4}{5}$ in $(1-x)^{1/2}$
17. [T] $\sqrt{3} = \frac{3}{\sqrt{3}}$ using $x = \frac{2}{3}$ in $(1-x)^{1/2}$
18. [T] $\sqrt{6}$ using $x = \frac{5}{6}$ in $(1-x)^{1/2}$
19. Integrate the binomial approximation of $\sqrt{1-x}$ to find an approximation of $\int_0^x \sqrt{1-t} dt$.
20. [T] Recall that the graph of $\sqrt{1-x^2}$ is an upper semicircle of radius 1. Integrate the binomial approximation of $\sqrt{1-x^2}$ up to order 8 from $x = -1$ to $x = 1$ to estimate $\frac{\pi}{2}$.

In the following exercises, use the expansion $(1+x)^{1/3} = 1 + \frac{1}{3}x - \frac{1}{9}x^2 + \frac{5}{81}x^3 - \frac{10}{243}x^4 + \dots$ to write the first five terms (not necessarily a quartic polynomial) of each expression.

21. $(1+4x)^{1/3}; a = 0$
22. $(1+4x)^{4/3}; a = 0$
23. $(3+2x)^{1/3}; a = -1$
24. $(x^2+6x+10)^{1/3}; a = -3$
25. Use $(1+x)^{1/3} = 1 + \frac{1}{3}x - \frac{1}{9}x^2 + \frac{5}{81}x^3 - \frac{10}{243}x^4 + \dots$ with $x = 1$ to approximate $2^{1/3}$.
26. Use the approximation $(1-x)^{2/3} = 1 - \frac{2x}{3} - \frac{x^2}{9} - \frac{4x^3}{81} - \frac{7x^4}{243} - \frac{14x^5}{729} + \dots$ for $|x| < 1$ to approximate $2^{1/3} = 2 \cdot 2^{-2/3}$.
27. Find the 25th derivative of $f(x) = (1+x^2)^{13}$ at $x = 0$.
28. Find the 99th derivative at $x = 0$ of $f(x) = (1+x^4)^{25}$.

In the following exercises, find the Maclaurin series of each function.

29. $f(x) = xe^{2x}$
30. $f(x) = 2^x$
31. $f(x) = \frac{\sin x}{x}$
32. $f(x) = \frac{\sin(\sqrt{x})}{\sqrt{x}}$, where $x > 0$.
33. $f(x) = \sin(x^2)$
34. $f(x) = e^{x^3}$
35. $f(x) = \cos^2 x$ using the identity $\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos(2x)$
36. $f(x) = \sin^2 x$ using the identity $\sin^2 x = \frac{1}{2} - \frac{1}{2} \cos(2x)$

In the following exercises, find the Maclaurin series of $F(x) = \int_0^x f(t) dt$ by integrating the Maclaurin series of f term by term. If f is not strictly defined at zero, you may substitute the value of the Maclaurin series at zero.

37. $F(x) = \int_0^x e^{-t^2} dt; f(t) = e^{-t^2} = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{n!}$

38. $F(x) = \tan^{-1} x$; $f(t) = \frac{1}{1+t^2} = \sum_{n=0}^{\infty} (-1)^n t^{2n}$
39. $F(x) = \tanh^{-1} x$; $f(t) = \frac{1}{1-t^2} = \sum_{n=0}^{\infty} t^{2n}$
40. $F(x) = \sin^{-1} x$; $f(t) = \frac{1}{\sqrt{1-t^2}} = \sum_{k=0}^{\infty} \frac{(2k)!}{4^k (k!)^2} t^{2k}$
41. $F(x) = \int_0^x \frac{\sin t}{t} dt$; $f(t) = \frac{\sin t}{t} = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{(2n+1)!}$
42. $F(x) = \int_0^x \cos(\sqrt{t}) dt$; $f(t) = \sum_{n=0}^{\infty} (-1)^n \frac{t^n}{(2n)!}$
43. $F(x) = \int_0^x \frac{1-\cos t}{t^2} dt$; $f(t) = \frac{1-\cos t}{t^2} = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n}}{(2n+2)!}$
44. $F(x) = \int_0^x \frac{\ln(1+t)}{t} dt$; $f(t) = \sum_{n=0}^{\infty} (-1)^n \frac{t^n}{n+1}$

In the following exercises, compute at least the first three nonzero terms (not necessarily a quadratic polynomial) of the Maclaurin series of f .

45. $f(x) = \sin(x + \frac{\pi}{4}) = \sin x \cos(\frac{\pi}{4}) + \cos x \sin(\frac{\pi}{4})$
46. $f(x) = \tan x$
47. $f(x) = \ln(\cos x)$
48. $f(x) = e^x \cos x$
49. $f(x) = e^{\sin x}$
50. $f(x) = \sec^2 x$
51. $f(x) = \tanh x$
52. $f(x) = \frac{\tan \sqrt{x}}{\sqrt{x}}$ (see expansion for $\tan x$)

In the following exercises, find the radius of convergence of the Maclaurin series of each function.

53. $\ln(1+x)$
54. $\frac{1}{1+x^2}$
55. $\tan^{-1} x$
56. $\ln(1+x^2)$
57. Find the Maclaurin series of $\sinh x = \frac{e^x - e^{-x}}{2}$.
58. Find the Maclaurin series of $\cosh x = \frac{e^x + e^{-x}}{2}$.
59. Differentiate term by term the Maclaurin series of $\sinh x$ and compare the result with the Maclaurin series of $\cosh x$.
60. [T] Let $S_n(x) = \sum_{k=0}^n (-1)^k \frac{x^{2k+1}}{(2k+1)!}$ and $C_n(x) = \sum_{k=0}^n (-1)^k \frac{x^{2k}}{(2k)!}$ denote the respective Maclaurin polynomials of degree $2n+1$ of $\sin x$ and degree $2n$ of $\cos x$. Plot the errors $\frac{S_n(x)}{C_n(x)} - \tan x$ for $n = 1, \dots, 5$ and compare them to $x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} - \tan x$ on $(-\frac{\pi}{4}, \frac{\pi}{4})$.
61. Use the identity $2 \sin x \cos x = \sin(2x)$ to find the power series expansion of $\sin^2 x$ at $x = 0$. (Hint: Integrate the Maclaurin series of $\sin(2x)$ term by term.)
62. If $y = \sum_{n=0}^{\infty} a_n x^n$, find the power series expansions of xy' and $x^2 y''$.

63. [T] Suppose that $y = \sum_{k=0}^{\infty} a_k x^k$ satisfies $y' = -2xy$ and $y(0) = 1$. Show that $a_{2k+1} = 0$ for all k and that $a_{2k+2} = \frac{a_{2k}}{k+1}$. Plot the partial sum S_{20} of y on the interval $[-4, 4]$.
64. [T] Suppose that a set of standardized test scores is normally distributed with mean $\mu = 100$ and standard deviation $\sigma = 10$. Set up an integral that represents the probability that a test score will be between 90 and 110 and use the integral of the degree 10 Maclaurin polynomial of $\frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ to estimate this probability.
65. [T] Suppose that a set of standardized test scores is normally distributed with mean $\mu = 100$ and standard deviation $\sigma = 10$. Set up an integral that represents the probability that a test score will be between 70 and 130 and use the integral of the degree 50 Maclaurin polynomial of $\frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ to estimate this probability.
66. [T] Suppose that $\sum_{n=0}^{\infty} a_n x^n$ converges to a function $f(x)$ such that $f(0) = 1$, $f'(0) = 0$, and $f''(x) = -f(x)$. Find a formula for a_n and plot the partial sum S_N for $N = 20$ on $[-5, 5]$.
67. [T] Suppose that $\sum_{n=0}^{\infty} a_n x^n$ converges to a function $f(x)$ such that $f(0) = 0$, $f'(0) = 1$, and $f''(x) = -f(x)$. Find a formula for a_n and plot the partial sum S_N for $N = 10$ on $[-5, 5]$.
68. Suppose that $\sum_{n=0}^{\infty} a_n x^n$ converges to a function y such that $y'' - y' + y = 0$ where $y(0) = 1$ and $y'(0) = 0$. Find a formula that relates a_{n+2} , a_{n+1} , and a_n and compute $a_0, \dots, a_5$.
69. Suppose that $\sum_{n=0}^{\infty} a_n x^n$ converges to a function y such that $y'' - y' + y = 0$ where $y(0) = 0$ and $y'(0) = 1$. Find a formula that relates a_{n+2} , a_{n+1} , and a_n and compute $a_1, \dots, a_5$.

The error in approximating the integral $\int_a^b f(t) dt$ by that of a Taylor approximation $\int_a^b P_n(t) dt$ is at most $\int_a^b |R_n(t)| dt$. In the following exercises, the Taylor remainder estimate $|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}$ guarantees that the integral of the Taylor polynomial of the given order approximates the integral of f with an error less than $\frac{1}{10}$.

- (a) Evaluate the integral of the appropriate Taylor polynomial and verify that it approximates the CAS value with an error less than $\frac{1}{100}$.
- (b) Compare the accuracy of the polynomial integral estimate with the remainder estimate.

70. [T] $\int_0^{\pi} \frac{\sin t}{t} dt$; $P_8(t) = 1 - \frac{t^2}{3!} + \frac{t^4}{5!} - \frac{t^6}{7!} + \frac{t^8}{9!}$ (You may assume that the absolute value of the ninth derivative of $\frac{\sin t}{t}$ is bounded by 0.1.)
71. [T] $\int_0^2 e^{-x^2} dx$; $P_{22}(x) = 1 - x^2 + \frac{x^4}{2} - \frac{x^6}{3!} + \dots - \frac{x^{22}}{11!}$ (You may assume that the absolute value of the 23rd derivative of e^{-x^2} is less than 2×10^{14} .)

The following exercises deal with Fresnel integrals.

72. The Fresnel integrals are defined by $C(x) = \int_0^x \cos(t^2) dt$ and $S(x) = \int_0^x \sin(t^2) dt$. Compute the power series of $C(x)$ and $S(x)$ and plot the sums $C_N(x)$ and $S_N(x)$ of the first $N = 50$ nonzero terms on $[0, 2\pi]$.
73. [T] The Fresnel integrals are used in design applications for roadways and railways and other applications because of the curvature properties of the curve with coordinates $(C(t), S(t))$. Plot the curve (C_{50}, S_{50}) for $0 \leq t \leq 2\pi$, the coordinates of which were computed in the previous exercise.

74. Estimate $\int_0^{1/4} \sqrt{x-x^2} dx$ by approximating $\sqrt{1-x}$ using the binomial approximation $1 - \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \frac{5x^4}{128} + \frac{7x^5}{256}$.
75. [T] Use Newton's approximation of the binomial $\sqrt{1-x^2}$ to approximate π as follows. The circle centered at $(\frac{1}{2}, 0)$ with radius $\frac{1}{2}$ has upper semicircle $y = \sqrt{x}\sqrt{1-x}$. The sector of this circle bounded by the x -axis between $x = 0$ and $x = \frac{1}{2}$ and by the line joining $(\frac{1}{4}, \frac{\sqrt{3}}{4})$ corresponds to $\frac{1}{6}$ of the circle and has area $\frac{\pi}{24}$. This sector is the union of a right triangle with height $\frac{\sqrt{3}}{4}$ and base $\frac{1}{4}$ and the region below the graph between $x = 0$ and $x = \frac{1}{4}$. To find the area of this region you can write $y = \sqrt{x}\sqrt{1-x} = \sqrt{x} \times (\text{binomial expansion of } \sqrt{1-x})$ and integrate term by term. Use this approach with the binomial approximation from the previous exercise to estimate π .
76. Use the approximation $T \approx 2\pi\sqrt{\frac{L}{g}}(1 + \frac{k^2}{4})$ to approximate the period of a pendulum having length 10 meters and maximum angle $\theta_{\max} = \frac{\pi}{6}$ where $k = \sin(\frac{\theta_{\max}}{2})$. Compare this with the small angle estimate $T \approx 2\pi\sqrt{\frac{L}{g}}$.
77. Suppose that a pendulum is to have a period of 2 seconds and a maximum angle of $\theta_{\max} = \frac{\pi}{6}$. Use $T \approx 2\pi\sqrt{\frac{L}{g}}(1 + \frac{k^2}{4})$ to approximate the desired length of the pendulum. What length is predicted by the small angle estimate $T \approx 2\pi\sqrt{\frac{L}{g}}$?
78. Evaluate $\int_0^{\pi/2} \sin^4 \theta d\theta$ in the approximation $T = 4\sqrt{\frac{L}{g}} \int_0^{\pi/2} (1 + \frac{1}{2}k^2 \sin^2 \theta + \frac{3}{8}k^4 \sin^4 \theta + \dots) d\theta$ to obtain an improved estimate for T .
79. [T] An equivalent formula for the period of a pendulum with amplitude $\theta_{\max}$ is $T(\theta_{\max}) = 2\sqrt{2}\sqrt{\frac{L}{g}} \int_0^{\theta_{\max}} \frac{d\theta}{\sqrt{\cos \theta - \cos(\theta_{\max})}}$ where L is the pendulum length and g is the gravitational acceleration constant. When $\theta_{\max} = \frac{\pi}{3}$ we get $\frac{1}{\sqrt{\cos t - 1/2}} \approx \sqrt{2}(1 + \frac{t^2}{2} + \frac{t^4}{3} + \frac{181t^6}{720})$. Integrate this approximation to estimate $T(\frac{\pi}{3})$ in terms of L and g . Assuming $g = 9.806$ meters per second squared, find an approximate length L such that $T(\frac{\pi}{3}) = 2$ seconds.